

## 1. What is Hacking?

**Hacking** is the process of identifying weaknesses in a system or network to gain unauthorized access or control over its data, resources, or systems.

 Originally, hacking was used positively for innovation and problem-solving. Today, it's often associated with **cybercrimes**, although not all hacking is illegal.

### Examples:

- Bypassing login authentication
- Injecting malicious code into websites
- Gaining root access to a server

## 2. Types of Hackers

| Type of Hacker          | Description                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| White Hat               | Ethical hackers who test systems to improve security (e.g., security auditors).                                          |
| Black Hat               | Malicious hackers who exploit systems for personal gain or damage.                                                       |
| Grey Hat                | Hackers who may violate laws but do not have malicious intent (e.g., finding a bug without permission and reporting it). |
| Script Kiddies          | Inexperienced users using tools/scripts without understanding them.                                                      |
| Hacktivists             | Hackers motivated by political or social causes (e.g., Anonymous group).                                                 |
| State-Sponsored Hackers | Hired by governments to infiltrate or defend against cyber attacks.                                                      |

## 3. Phases of Ethical Hacking

Ethical hacking follows a **step-by-step methodology**, often called the **penetration testing lifecycle**:

| Phase                 | Description                                                           |
|-----------------------|-----------------------------------------------------------------------|
| 1. Reconnaissance     | Gathering information about the target (passive/active).              |
| 2. Scanning           | Identifying live hosts, open ports, services, and vulnerabilities.    |
| 3. Gaining Access     | Exploiting vulnerabilities to enter the system (e.g., SQL injection). |
| 4. Maintaining Access | Establishing persistence through backdoors or rootkits.               |
| 5. Clearing Tracks    | Erasing logs, clearing history to avoid detection.                    |
| 6. Reporting          | Documenting findings and recommending security improvements.          |

## 4. Legal Aspects & Cyber Laws (India & Global)

Ethical hacking must be done **with proper permission**. Performing unauthorized tests is illegal under cyber laws.

### Legal Boundaries:

- Always have **written consent** before testing.
- Follow a **Non-Disclosure Agreement (NDA)**.
- Respect **data privacy** and **integrity**.

### Indian Laws (IT Act 2000):

- **Section 66** – Hacking, data theft: 3 years jail or ₹5 lakh fine.
- **Section 43** – Unauthorized access, data damage.
- **Section 66F** – Cyber terrorism (severe punishments).

### Global Laws:

- **Computer Fraud and Abuse Act (USA)**
- **GDPR (EU)** – For handling personal data
- **PCI-DSS** – For organizations dealing with card payments
- **Important:** Unauthorized hacking = Cyber Crime!

## 5. Penetration Testing vs Ethical Hacking

| Criteria          | Ethical Hacking                                | Penetration Testing                               |
|-------------------|------------------------------------------------|---------------------------------------------------|
| <b>Definition</b> | Broad practice of testing and securing systems | Specific process of simulating real-world attacks |
| <b>Objective</b>  | Improve overall security posture               | Identify exploitable vulnerabilities              |
| <b>Scope</b>      | Includes policies, social engineering, etc.    | Focused on technical systems only                 |
| <b>Approach</b>   | Continuous security improvement                | Often periodic or project-based                   |
| <b>Example</b>    | Security audit for entire organization         | Testing a web app for SQL injection               |

✅ **Penetration Testing is a subset of Ethical Hacking.**